

BSc (Hons) Computing - Coventry University
Higher National Diploma in Software Engineering - NIBM



Data Warehousing and Business Intelligence

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6.1 Introduction to Data Warehousing

Learning Objectives



Understand the key concepts of data warehousing -
Definition, purpose, and benefits.

Explore the evolution of Decision Support Systems (DSS)
- *From early data systems to modern BI tools.*

Recognize the importance of data warehousing -
*Its role in enhancing business intelligence and
decision-making.*

Business Intelligence



Business Intelligence (BI) is a set of tools, technologies, and practices that transform raw data into meaningful and actionable information, empowering organizations to make informed decisions, optimize processes, and drive better business outcomes.

BI involves the collection, integration, analysis, and presentation of data from various sources, enabling users to gain insights into their business performance, identify trends and patterns, and develop effective strategies for growth and improvement.

What is Data Warehousing?



- Centralized repository that stores and manages large volumes of data.
- Collects data from various sources, such as operational systems, external databases, and other internal systems.
- Data is structured, cleaned, and stored for analysis and reporting purposes.
- Supports decision-making by providing historical data insights.
- Primarily used for business intelligence, trend analysis, and reporting.

Example:

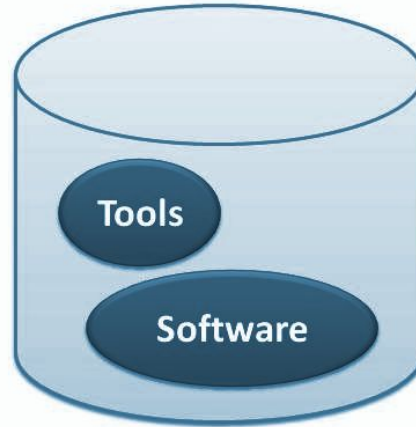
A retail company stores all sales data in a data warehouse to analyze yearly trends and customer behavior.

What makes up a Data Warehouse?

Data from different Sources



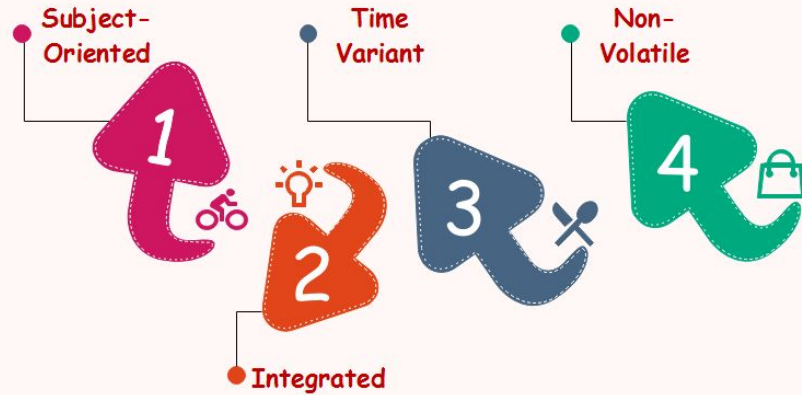
Data Warehouse



Use of data for



Data Warehouse - Essential Characteristics



Non-Volatile

Data is stable and not frequently modified or deleted, ensuring historical integrity.

Subject-Oriented

Organized around key business subjects (e.g., customers, sales, finance).

Time-Variant

Stores historical data over a long period for trend analysis (e.g., daily, weekly, yearly snapshots).

Integrated

Combines data from multiple sources (databases, applications) into a unified format.

Data Warehouse - Components



Front-End Tools

- Dashboards, reports, and visualization tools for analyzing and presenting data.
- Examples: Microsoft Power BI, Oracle BI, Tableau

Metadata

- Information about data (e.g., source, usage, definitions).
- Helps in understanding, managing, and utilizing the data.

ETL Process

- **Extract:** Data is gathered from various sources.
- **Transform:** Data is cleaned, formatted, and structured.
- **Load:** Transformed data is loaded into the warehouse.

Data Storage

- Centralized repository for organized and structured data.
- Supports query and reporting tools.

Discussion

The difference - Database vs. Data Warehouse?

Databases



OLTP - Online Transaction Processing

- Designed for real-time operations and transactions (e.g., banking, retail systems).
- Focus: Quick read/write operations (e.g., inserts, updates, deletes).
- Data: Current, detailed, and transactional.

Data Warehouses



OLAP - Online Analytical Processing

- **Optimized for analyzing large volumes of historical data.**
- **Focus: Complex queries and data analysis (e.g., trends, forecasts).**
- **Data: Aggregated, historical, and subject-oriented.**

Discussion

... Continued.

The difference - Database vs. Data Warehouse?

Database:

Supports day-to-day operations, fast transactions.

Data Warehouse:

Supports business intelligence, complex queries, and analysis.

Discussion

Benefits of Data Warehousing?

Data Warehousing

- Benefits



Improved Decision-Making

- Provides accurate, consistent data to support strategic decisions.

Historical Data Analysis

- Enables in-depth analysis of trends over time (e.g., sales growth, customer behavior).

Enhanced Reporting and Querying

- Facilitates complex queries and reports without impacting operational systems.

Data Consolidation

- Integrates data from multiple sources, offering a single source of truth for reporting.

Supports Business Intelligence (BI) Applications

- Acts as a foundation for BI tools, enabling data visualization and predictive analytics.

Discussion

... Continued.

Benefits of Data Warehousing?

Example:

A healthcare organization analyzes historical patient data to identify treatment effectiveness and improve patient care.

Informed Decision-Making in Business

Competitive Advantage: Making timely, data-backed decisions helps businesses stay ahead in competitive markets. Helps identify new opportunities, predict trends, and respond effectively to market changes.

Risk Reduction: Informed decisions reduce uncertainties by providing insights into potential risks. Businesses can anticipate challenges, minimize losses, and avoid costly mistakes.

Improved Operational Efficiency: Better decision-making streamlines processes, optimizes resource allocation, and boosts productivity. Leads to more efficient workflows, minimizing waste and improving profit margins.

Informed Decision-Making in Business

... Continued.

Enhanced Customer Satisfaction: Data-driven insights help in understanding customer preferences and behaviors. Allows businesses to make customer-centric decisions, improving service quality and loyalty.

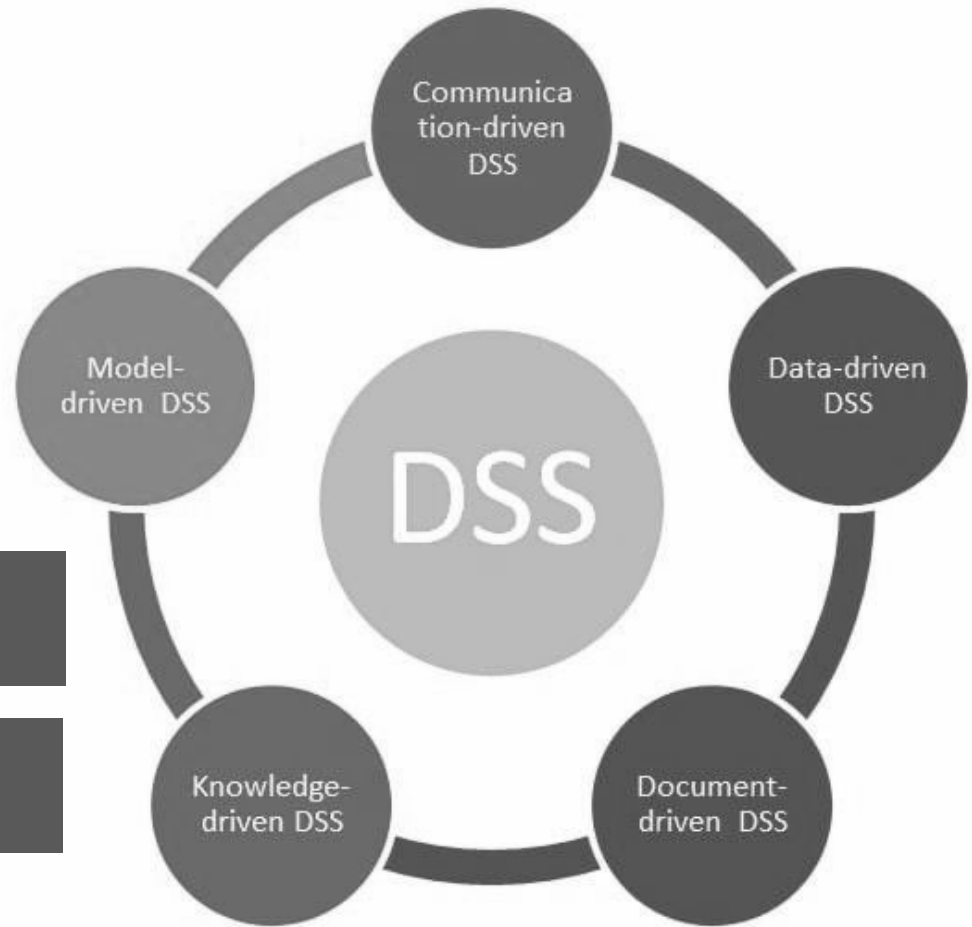
Resource Optimization: Informed decisions ensure resources are allocated where they are most needed, maximizing return on investment (ROI). Avoids unnecessary expenditure and enhances overall financial health.

Long-Term Success and Stability: Consistently making well-informed decisions builds a strong foundation for sustainable growth. Aligns strategic goals with actionable insights, supporting long-term planning and stability.

Decision Support Systems (DSS)

Interactive software designed to aid decision-making processes within organizations.

- Combines data, sophisticated models, and user-friendly interfaces for analysis.
- Helps users analyze data and make informed decisions.



Questions and Discussion

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